MSC MARRIAGE AND FAMILY THERAPY

LECTURE LIBRARY VOLUME 21 OF 28

MFT 721

Research Methodology

14 teaching weeks 2 practicals 2 figures

SEMESTER THREE

1.71 units Prerequisite: MFT 705

> This is the advanced methods course that turns a proposal into a thesis.

One of 28 self-contained course books in the MSc Marriage and Family Therapy lecture library: a full taught semester per volume, with cases, practicals, assessment briefs, glossaries and model answers.

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MFT 721 Research Methodology

MSc Marriage and Family Therapy Semester Three 1.71 Units 14 teaching weeks Prerequisite: MFT 705

This volume is one of 28 course books that together cover the whole MSc Marriage and Family Therapy taught programme. Each book is written to carry a 14 16 week semester on its own: every week pairs a core idea with the practice that follows from it, a worked case, and a reflective task, and each book closes with an integration chapter, a glossary and a self-assessment paper with model answers.

The books are designed to be read straight through before the semester begins, revisited week by week, and used again in revision. Cross-references point to the other volumes where a topic is treated in more depth.

How to cite this volume

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A note on the case material

All families, couples and individuals in this book are fictional composites. They are written to teach pattern recognition, not to describe any real person. Nothing here replaces supervised clinical practice, professional registration, legal advice or medical care; legal content is orientation only and must be checked against the law of the relevant jurisdiction.

The lecture library

Twenty-eight self-contained course books carry the taught programme: eight in Semester One, seven in Semester Two, seven in Semester Three including the thesis, and six electives. The volume you are holding is marked in the margin.

Code	Course	Semester	Units
MFT 701	Fundamentals of Marriage and Family Therapy	One	1.71
MFT 702	Theories of Family Therapy	One	1.71
MFT 703	Human Development Across the Lifespan	One	1.71
MFT 704	Ethical and Legal Issues in Therapy	One	1.71
MFT 705	Research Methods in Marriage and Family Therapy	One	1.71
MFT 706	Clinical Practice and Techniques	One	1.71
MFT 707	Assessment and Diagnosis in Family Therapy	One	1.71
MFT 708	Cultural Competence in Therapy	One	1.71
MFT 709	Group Therapy Techniques	Two	1.71
MFT 710	Couples Therapy	Two	1.71
MFT 711	Child and Adolescent Therapy	Two	1.71
MFT 712	Crisis Intervention Strategies	Two	1.71
MFT 713	Substance Abuse and Family Dynamics	Two	1.71
MFT 714	Trauma-Informed Care	Two	1.71
MFT 715	Family Systems Theory	Two	1.71
MFT 716	Communication Skills in Therapy	Three	1.71
MFT 717	Psychopathology in Families	Three	1.71
MFT 718	Intimacy and Relationship Dynamics	Three	1.71
MFT 719	Family Life Cycle and Transitions	Three	1.71
MFT 720	Supervision and Consultation in Therapy	Three	1.71
MFT 721	**Research Methodology**	Three	1.71
MFT 722	Professional Development and Self-Care	Electives	1.71
MFT 723	Special Populations in Therapy	Electives	1.71
MFT 724	Family Law and Advocacy	Electives	1.71
MFT 725	Integrative Approaches in Therapy	Electives	1.71
MFT 726	Neurobiology of Relationships	Electives	1.71
MFT 727	Play Therapy Techniques	Electives	1.71
MFT 770	Master Thesis	Three	1.71

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Course leader's summary. This is the advanced methods course that turns a proposal into a thesis. You will work with real data, choose and justify analytic strategies, defend your design against examiners, and learn to write findings that practitioners will actually read.

How to use this book

Every week's task contributes directly to your thesis. Bring your own dataset, or use the teaching dataset supplied with the course. Do not read statistics passively; run them, break them, and interpret them badly at least once.

1. Course description

The course covers advanced design (longitudinal, dyadic, single-case, participatory), analysis (regression, multilevel modelling, dyadic models, thematic and framework analysis, mixed-methods integration), measurement and psychometric evaluation, systematic reviewing, data management and reproducibility, writing for publication, and defence of a study.

2. Learning outcomes

#	Outcome	Assessed
LO1	Select and justify a design for a complex relational question	Thesis proposal
LO2	Conduct and interpret appropriate analyses for relational data	Analysis portfolio
LO3	Evaluate the psychometric quality of a measure in a new population	Psychometric report
LO4	Integrate quantitative and qualitative findings coherently	Mixed-methods report
LO5	Defend design decisions against alternative explanations	Viva
LO6	Write a publishable findings chapter	Thesis chapter
LO7	Manage data ethically, securely and reproducibly	Data plan and audit

3. Assessment

Component	Weight	Due
Analysis portfolio (4 exercises)	30%	Weeks 5, 8, 11, 13
Psychometric evaluation report (1,500 words)	20%	Week 9
Mixed-methods integration report (2,000 words)	20%	Week 12
Thesis methodology chapter + viva	30%	Week 15

4. The fourteen-week plan

Week	Topic	Task
1	From proposal to protocol; pre-registration and its limits	Protocol draft
2	Design for relational questions	Design decision log
3	Sampling, power and feasibility in clinical settings	Power calculation
4	Data preparation and the analysis plan	Data cleaning log
5	**Analysis 1:** regression and its assumptions	Run and interpret
6	Multilevel models for nested family data	Model comparison
7	Dyadic models: APIM in practice	APIM exercise
8	**Analysis 2:** thematic and framework analysis	Coding audit
9	Psychometrics: reliability, factor structure, invariance	Report
10	Systematic review and meta-analysis basics	Search and appraise
11	**Analysis 3:** single-case and small-N designs	SCED exercise
12	Mixed methods and integration	Joint display
13	**Analysis 4:** robustness, missing data, common pitfalls	Sensitivity analysis
14	Writing, publishing and defending	Chapter and viva prep

Week 1 Protocol and pre-registration

The protocol is the study's constitution: question, design, sample, measures, analysis plan, ethics, and timeline. Write it before you collect data; revise it with dated amendments if you must change course.

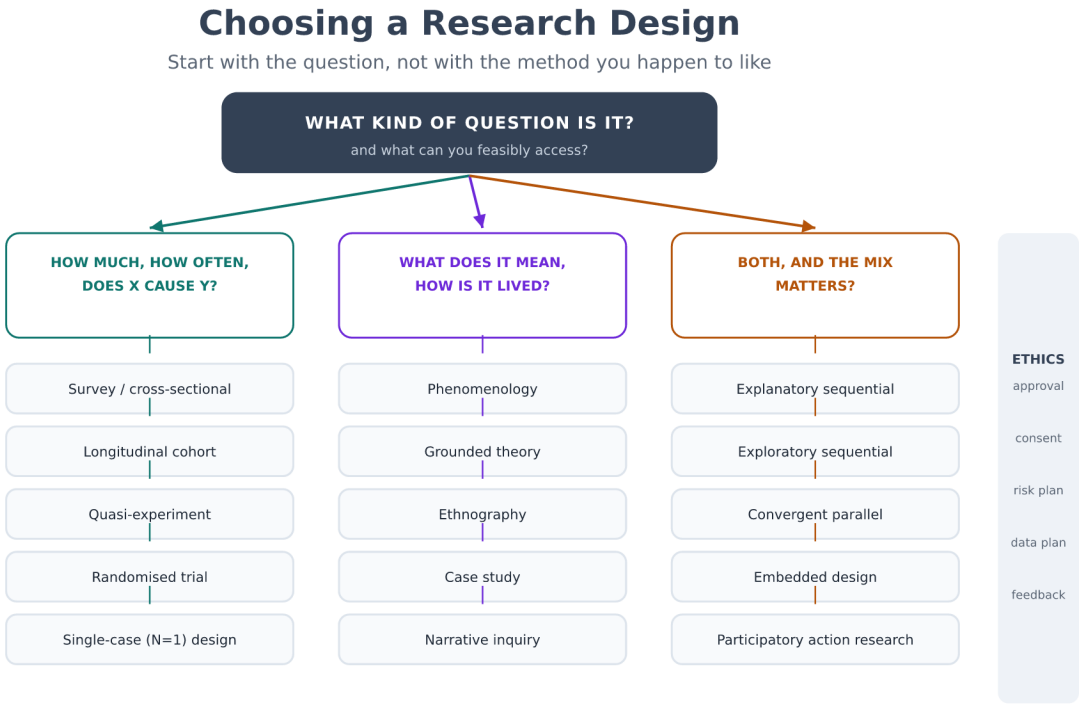
Pre-registration (e.g. on OSF) commits you publicly to a plan, which reduces the risk of unconscious analytical flexibility ("p-hacking"). It is standard in trials and increasingly expected elsewhere.

Honest limits. Exploratory qualitative work and community-based participatory research are often incompatible with rigid pre-registration. Declare your approach and be transparent about decisions made during the study.

A live analysis log one page per week, dated is the single most useful document in a thesis. It records what you did, what you decided, and why.

> **TASK**
> Draft a two-page protocol for your thesis. Include one paragraph on what you will do if recruitment is half of what you expect.

Week 2 Design for relational questions



NOTE

In MFT the unit of analysis is the relationship, not only the individual — which creates real statistical problems: partners are non-independent. Plan for it (dyadic data analysis, actor–partner models, or a design that samples one person per household).

Figure 2.1 Choosing a design from the question

Four designs especially suited to family questions.

Design	Question	Strength	Weakness
Longitudinal panel	Does X predict change in Y?	Temporal ordering	Attrition, cost
Dyadic (APIM)	How do partners affect each other?	Separates actor and partner effects	Requires both partners
Single-case experimental (SCED)	Does this intervention work for this family?	Clinical realism, causal inference within case	Generalisation
Participatory action research	What would improve this community's experience?	Validity, ethics, impact	Time, power negotiation

The design decision log. For each choice, record: the alternative considered, the reason for rejection, and the threat it leaves unresolved. Examiners reward this; it shows you know what your design cannot do.

Week 3 Sampling, power and feasibility

Power. Do the calculation before you collect (GPower, R's `pwr`, or simulation). For dyadic and multilevel designs, power depends on the number of clusters* (couples, families), not the number of individuals.

Practical realities in clinical settings.

- Referral rates: how many families does the clinic actually see per week?
- Eligibility: what proportion will meet your criteria?
- Consent: what proportion will agree? (Often 30 60% in clinical samples.)
- Attrition: assume 20 30% at follow-up.
- Seasonality: school terms, holidays, religious calendars.

The feasibility sentence. "To recruit 60 couples in six months, I need 4 eligible referrals per week with a 60% consent rate; the clinic receives 9 per week; two colleagues have agreed to refer."

Week 4 Data preparation

The unglamorous work that determines everything.

1. **Codebook** every variable, its label, its values, its missingness convention.
2. **Raw data frozen** never edit the original file after data entry.
3. **Cleaning log** every decision (a value of 99 recoded to missing; one duplicate removed).
4. **Missing data** describe the pattern; distinguish missing at random from missing not at random; consider multiple imputation rather than listwise deletion.
5. **Reverse scoring** check; a mis-scored subscale invalidates the results.
6. **Outliers** inspect, do not automatically delete.

> **TASK**
> Produce a codebook and cleaning log for your dataset (or the teaching dataset).

Week 5 Regression

What regression does. Estimates the association between predictors and an outcome, adjusting for other predictors, and expresses it in the outcome's units.

Assumptions to check. Linearity, independence of errors, homoscedasticity, normality of residuals (for inference), and no severe multicollinearity. Report the checks; do not just assert them.

Reading output. Coefficient (direction, size, units), standard error, confidence interval, p-value, R , and model fit. Report effect sizes with confidence intervals, not only p-values.

Clinical translation. "Each additional point on the criticism scale was associated with a 0.3-point lower satisfaction score (95% CI 0.5 to 0.1), holding depression constant." That sentence is worth more than a table of asterisks.

Week 6 Multilevel models

The problem. Individuals nested in couples, couples nested in therapists, repeated measures nested in people. Standard regression assumes independence and will produce confidence intervals that are too narrow.

The multilevel solution. Model the nesting explicitly:

- **Fixed effects** the average relationship across the sample.
- **Random effects** how intercepts (and slopes) vary across clusters.
- **ICC** the proportion of variance attributable to the cluster. A high ICC means the clustering matters greatly and cannot be ignored.

Model-building sequence. Null model (ICC) add level-1 predictors add level-2 predictors test random slopes compare with likelihood ratio tests or information criteria (AIC/BIC).

Report. Always report the ICC, the number of clusters, and the estimation method.

Week 7 Dyadic models

APIM in four numbers.

- **Actor effect (a):** my predictor on my outcome.
- **Partner effect (p):** my predictor on my partner's outcome.
- **Distinguishability:** whether partners are distinguishable (e.g. by gender or role) and whether effects can be constrained equal.
- **K:** the correlation between partners' residuals, which models the non-independence.

Interpretation example. In a study of 80 couples, criticism predicted lower satisfaction for the person giving it (actor effect) more strongly than for the person receiving it (partner effect). Clinical implication: the critic is also being harmed, which is a more useful reframe than "stop criticising your partner".

Software. R (`lme4`, `nlme`, `ape`), Mplus, SPSS MIXED.

Week 8 Qualitative analysis

Two workable approaches.

Approach	Logic	Output
Reflexive thematic analysis (Braun & Clarke)	Themes constructed from coding, with the analyst's role explicit	Rich thematic account

Approach	Logic	Output
Framework analysis (Ritchie & Spencer)	Matrix of cases themes, often for applied policy questions	Structured matrix, comparable across cases

Rigour in practice.

- Keep a **coding audit trail**: initial codes, revisions, discarded codes and why.
- Use **analytic memos** after each transcript.
- **Second-coder check** on at least 20% of data with a documented discussion of divergence.
- **Member reflection** where appropriate (not to validate themes, but to deepen them).
- **Reflexive journal** on your own reactions and assumptions.

The common error is presenting themes as "findings that emerged" from data, as though the analyst were not present. Themes are constructed; say so.

Week 9 Psychometrics



Figure 9.1 Validity, reliability and their limits

A psychometric evaluation of a measure in your population.

1. **Internal consistency** Cronbach's (and , which is more appropriate for congeneric items).

2. **Factor structure** exploratory or confirmatory factor analysis; report fit indices (CFI, TLI, RMSEA, SRMR).
3. **Construct validity** correlations with related and unrelated measures.
4. **Measurement invariance** configural, metric, scalar, across groups (gender, language, age).
5. **Responsiveness** change scores in a treated sample.

A local validation study is a perfectly good thesis, and often more useful to the field than another outcome study using an unvalidated instrument.

> **REPORT (20%)**
> A 1,500-word psychometric evaluation of one instrument, with recommendations.

Week 10 Systematic review basics

Steps. Register the protocol (PROSPERO) define the question (PICO) search systematically (at least three databases; record the strategy) screen (two reviewers; report agreement) extract data assess risk of bias (e.g. Cochrane RoB 2, or CASP for qualitative) synthesise (narrative or meta-analytic) assess certainty (GRADE).

Meta-analysis, briefly. Pool effect sizes with a random-effects model unless there is a justification for fixed effects; report heterogeneity (I ,); explore it with subgroup analysis or meta-regression if there are enough studies; assess publication bias with a funnel plot and Egger's test, understanding their limitations.

What examiners look for. A search strategy that could be reproduced; a bias assessment that is not perfunctory; and a conclusion proportionate to the evidence.

Week 11 Single-case and small-N designs

Why they matter in family therapy. Clinical samples are small, families are heterogeneous, and randomised trials are expensive. Single-case designs give causal inference within a case.

Core designs.

Design	Structure	Inference
A B	Baseline, then intervention	Weak (confounded by time)
A B A	Baseline, intervention, withdrawal	Stronger if reversal occurs
Multiple baseline across behaviours	Staggered introduction	Strong; no withdrawal needed
Multiple baseline across participants	Staggered across families	Strong; realistic in clinics

Analysis. Visual analysis (level, trend, variability) plus a quantitative index (e.g. Tau-U) and, ideally, systematic replication across cases.

Reporting standard. Follow the SCED reporting guidelines (e.g. SCRIBE), including baseline stability and treatment fidelity.

Week 12 Mixed methods integration

The integration question. What does the combination tell you that neither strand could alone?

Joint display format.

Quantitative finding	Qualitative finding	Convergence / divergence	Interpretation
Attendance predicted outcome (= .32)	Families described "being known" by the therapist as decisive	Divergence in mechanism	Attendance may be a proxy for alliance
Drop-out 34%	Families cited transport and work shifts	Convergence	Structural barriers, not motivation

Avoid the two common failures. Reporting two parallel studies with no integration; and using qualitative data merely to illustrate quantitative findings.

> **REPORT (20%)**
> A 2,000-word integration report with at least three joint display rows.

Week 13 Robustness and pitfalls

Pitfalls you will meet.

Pitfall	What it looks like	Fix
p-hacking	Many tests, reporting the significant ones	Pre-specify; adjust; report all
HARKing	Hypothesis stated after results known	Register the hypothesis
Overfitting	Model too complex for the sample	Simpler model; cross-validation
Ignoring clustering	Treating partners as independent	Multilevel or dyadic models
Selective reporting	Reporting one measure, ignoring another	Report all pre-specified outcomes
Causal language	"Caused" from cross-sectional data	Use associational language

Sensitivity analysis. Run your main analysis again with a different reasonable decision (e.g. excluding an outlier, using a different missing-data approach) and report whether conclusions change. If they do, say so.

Week 14 Writing, publishing and defending

The findings chapter. Present, do not interpret (that is the discussion's job). Structure by question. Report descriptive statistics before inferential. Use tables for numbers and prose for meaning.

The discussion. What you found how it compares with the literature mechanisms limitations (specific, not generic) implications for practice and policy future research.

Defending. Prepare for four questions: (1) Why this design? (2) What are the threats to validity and how did you address them? (3) Why these analyses? (4) What can and cannot be concluded? Rehearse short answers.

Publishing. Choose a journal that publishes practitioner research; follow the author guidelines exactly; expect a rejection and a revision; respond to reviewers point by point.

Glossary

APIM Actor Partner Interdependence Model. **Cluster** a grouping unit (couple, family, therapist) whose members are not independent. **GRADE** a system for rating the certainty of evidence. **ICC** intraclass correlation; the share of variance due to clustering. **Measurement invariance** equivalence of a measure's functioning across groups. **Pre-registration** public commitment to a study plan before data collection. **SCED** single-case experimental design. **Sensitivity analysis** re-analysis under different assumptions to test robustness.

Self-assessment

Multiple choice

1. Treating partners as independent observations primarily affects: (a) external validity (b) standard errors and Type I error (c) reliability (d) sampling

2. The ICC indicates: (a) internal consistency (b) the proportion of variance due to clustering (c) effect size (d) power

3. A multiple baseline design is strong because: (a) it requires withdrawal (b) staggered introduction controls for time (c) it uses large samples (d) it is qualitative

4. Joint displays are used to: (a) present descriptive statistics (b) integrate quantitative and qualitative findings (c) assess bias (d) calculate power

5. Pre-registration mainly reduces: (a) attrition (b) analytical flexibility and p-hacking (c) cost (d) sample size

6. Measurement invariance concerns: (a) whether a measure behaves equivalently across groups (b) internal consistency only (c) sample size (d) randomisation

7. A sensitivity analysis tests: (a) whether conclusions hold under different assumptions (b) the sample's normality (c) the coding frame (d) the ethics

8. HARKing means: (a) hypothesising after results are known (b) random sampling (c) qualitative coding (d) data entry

9. In framework analysis the output is typically: (a) a case theme matrix (b) a regression table (c) a factor loading matrix (d) a funnel plot

10. The most useful document in a thesis is: (a) the title page (b) a dated analysis log (c) the abstract (d) the reference list

Answer key

1 b 2 b 3 b 4 b 5 b 6 a 7 a 8 a 9 a 10 b

Short answer

1. Explain why clustering matters for standard errors.
2. Distinguish actor and partner effects with an example.
3. List five elements of a systematic review protocol.
4. What is the difference between a sensitivity analysis and a subgroup analysis?
5. Name four threats to validity in a pre-post clinic study.

Essay (choose one, 2,000 words)

1. Defend (or attack) the claim that randomised trials are the gold standard for family therapy research.
2. Present a design for a dyadic study of criticism and satisfaction, justifying every choice.
3. Critically appraise a systematic review of family interventions, focusing on method and certainty.

Further reading

- Kenny, D. A., Kashy, D. A., & Cook, W. L. (2006). Dyadic Data Analysis. Guilford.
- Hox, J., Moerbeek, M., & van de Schoot, R. (2017). Multilevel Analysis (3rd ed.). Routledge.
- Braun, V., & Clarke, V. (2022). Thematic Analysis: A Practical Guide. Sage.
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- Kazdin, A. E. (2021). Single-Case Research Designs (3rd ed.). OUP.

Next: MFT 770 Master Thesis